



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Artificial Intelligence

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

UNIT-03: CLASSIFICATION AND PREDICTION

Classification:

<https://drive.google.com/file/d/1wSRvRWBfPtFVe0lMEMGJTaiO8WDENuuW/view?usp=sharing>

Issues Regarding Classification and Prediction

The main issues regarding classification and prediction in Data Warehousing and Mining (DWM) can be divided into challenges related to data preparation, the model's performance and characteristics, and comparison/evaluation methods

1) Data Cleaning – Data cleaning involves removing the noise and treatment of missing values. The noise is removed by applying smoothing techniques and the problem of missing values is solved by replacing a missing value with most commonly occurring value for that attribute.

2) Relevance Analysis – Database may also have the irrelevant attributes. Correlation analysis is used to know whether any two given attributes are related.

3) Data Transformation and reduction – The data can be transformed by any of the following methods.

- **Normalization** – The data is transformed using normalization. Normalization involves scaling all values for given attribute in order to make them fall within a small specified range. Normalization is used when in the learning step, the neural networks or the methods involving measurements are used.
- **Generalization** – The data can also be transformed by generalizing it to the higher concept. For this purpose, we can use the concept hierarchies. The data can also be modified by applying a higher idea to it.

Classification by Decision Tree:

<https://drive.google.com/file/d/1j-GutMxyOdNJqA82ERJV0l0vpENJT90O/view?usp=sharing>

Bayesian Classification:

Bayesian Classification is a probabilistic supervised machine learning method used in data mining to classify data points into predefined categories.

It operates on Bayes' Theorem, predicting the probability that a given data record belongs to a specific class, rather than just assigning a definitive label.

Types of Bayesian Classifiers

1) Bayes' Theorem

* Bayes' Theorem:

- Bayes' theorem is also known as Bayes' rule, Bayes' law or Bayesian reasoning.
- It describes the probability of an event, based on prior knowledge of conditions that might be related to the event.
- Bayes' theorem was named after the British mathematician Thomas Bayes.
- In probability theory it relates the conditional probability & marginal probabilities of two random events.
- Bayes' theorem allows updating the probability prediction of an event by observing new information of the real world.
- It is a way to calculate the value of $P(B|A)$ with the knowledge of $P(A|B)$

$$P(A|B) = \frac{\text{no. of times A \& B}}{\text{no. of times B}}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B) \cdot P(B) \text{ — (1)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A) = P(B|A) \cdot P(A) \text{ — (ii)}$$

from Equation (1) & (ii)

L.H.S are equal.

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$\text{So, } P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

Where, $P(A|B)$ = Conditional probability of A given B

$P(B|A)$ = Conditional probability of B given A

$P(A)$ = probability of event A

$P(B)$ = probability of event B

- $P(A|B)$ is known as posterior, which we need to calculate, and it will be read as probability of hypothesis A when we have occurred an evidence B.
- $P(B|A)$ is called as likelihood, in which we consider that hypothesis B is true, then we calculate the probability of evidence B.
- $P(A)$ is called the prior probability, probability of hypothesis before considering the evidence.
- $P(B)$ is called Marginal probability, pure probability of an evidence.

2) Naïve Bayes Classifier

* Naïve Bayes classifier

- Naïve Bayes algorithm is a Supervised Learning algorithm which is based on Baye's theorem and used for solving classification problems.
- It is mainly used in text classification that includes a high-dimensional training dataset.
- Naïve Bayes classifier is one of the simple and most effective classification algorithms, which helps in building the fast machine learning models that can make quick predictions.
- It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
- Some popular examples of Naïve Bayes Algorithm are spam filtration, Sentimental Analysis and classifying articles.
- It is called naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.
- for example: Such as if the fruit is identified on the bases of color, shape and taste then red, spherical and sweet fruit is recognized as an apples.
- Hence each features individually contributes to identify that it is an apple without depending on each other.
- It is called Bayes because it depends on the principle of Baye's Theorem.

Naive Bayes Theorem

$$\therefore P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where.

$P(A|B)$ = posterior probability or conditional probability of A given B

$P(B|A)$ = Likelihood or conditional probability of B given A

$P(A)$ = prior probability or probability of event A

$P(B)$ = Marginal probability or probability of event B

Rule Based Classification

* Rule-based classification :-

- Rule-based classifiers are just another type of classifier.
- Rule-based classifier makes use of a set of IF-THEN rules for classification. we can express a rule in the following form -

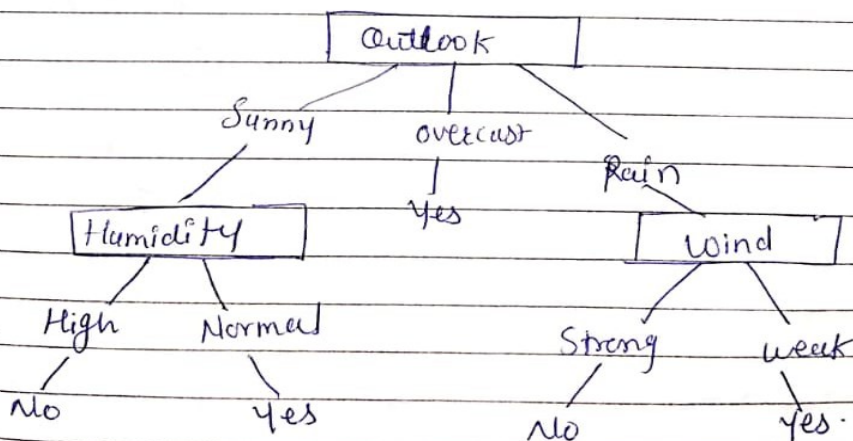
IF condition THEN conclusion.

- The IF part of the rule is called rule antecedent or precondition.
- The THEN part of the rule is called rule consequent.
- The consequent part consists of class prediction.

R1: IF age = 'youth' AND student = 'yes' THEN buys computer = 'yes'

Rule antecedent or pre-condition rule consequent.

- Example : Decision Tree for play Tennis



Rules from decision tree

R1: IF Outlook = 'Sunny' AND Humidity = 'High'
THEN play = 'No'

R2: IF Outlook = 'Sunny' AND Humidity = 'Normal'
THEN play = 'Yes'

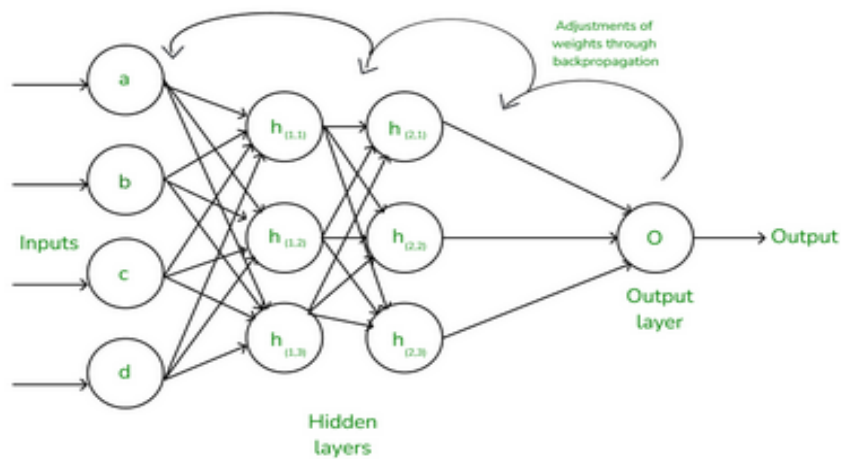
R3: IF Outlook = 'overcast' THEN play = 'yes'

R2: IF Outlook = 'Rain' AND Wind = 'Strong'
THEN play = 'No'

R2: IF Outlook = 'Rain' AND Wind = 'Weak'
THEN play = 'yes'

Classification by Back propagation

- Backpropagation is a technique used in deep learning to train artificial neural networks particularly feed-forward networks.
- It works iteratively to adjust weights and bias to minimize the cost function.
- In each epoch the model adapts these parameters reducing loss by following the error gradient.
- Backpropagation often uses optimization algorithms like gradient descent or stochastic gradient descent.
- The algorithm computes the gradient using the chain rule from calculus allowing it to effectively navigate complex layers in the neural network to minimize the cost function.
- Backpropagation (backward propagation of errors) is the key algorithm used to train artificial neural networks.
- It involves a two-step process: forward pass and backward pass.



Forward Pass

Input Data: The input data is fed into the network.

Propagation: The input data propagates through the network, layer by layer, to produce an output.

Loss Calculation: The output is compared with the actual target output to calculate the loss (error).

Backward Pass

Compute Gradients: The loss is propagated backward through the network to compute gradients of the loss with respect to each weight using the chain rule of calculus.

Update Weights: The weights are updated using gradient descent (or one of its variants) to minimize the loss.

The process is repeated for many iterations (epochs) until the network converges to a state where the loss is minimized.

Backpropagation Algorithm

Step 1: Inputs X , arrive through the pre-connected path.

Step 2: The input is modeled using true weights W . Weights are usually chosen randomly.

Step 3: Calculate the output of each neuron from the input layer to the hidden layer to the output layer.

Step 4: Calculate the error in the outputs

Backpropagation Error Actual Output - Desired Output

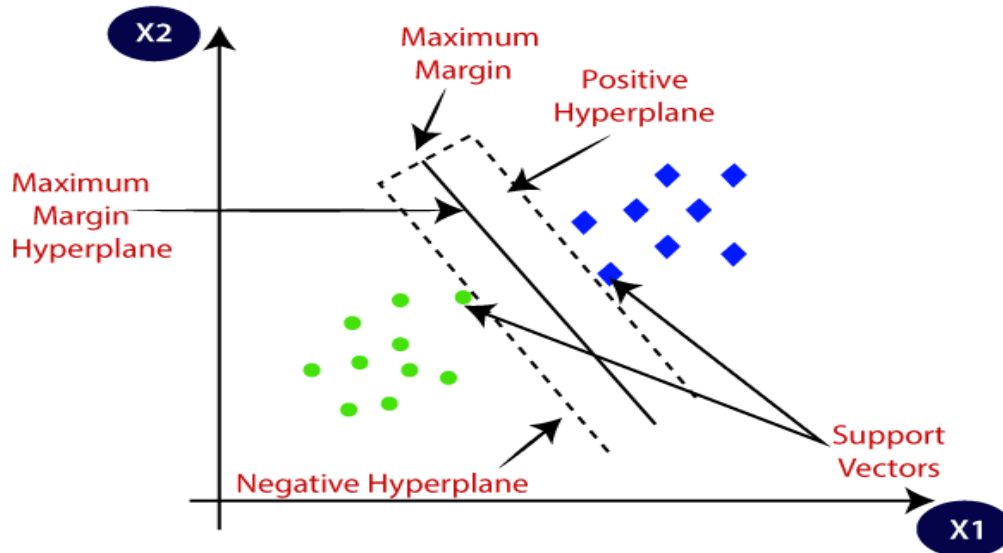
Step 5: From the output layer, go back to the hidden layer to adjust the weights to reduce the error.

Step 6: Repeat the process until the desired output is achieved.

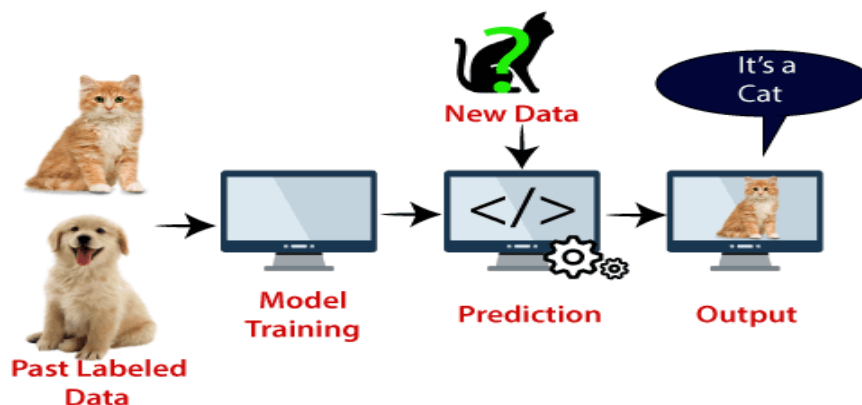
Support vector machines

- Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n -dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

- SVM chooses the extreme points/vectors that help in creating the hyperplane.
- These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine.
- SVM algorithm can be used for Face detection, image classification, text categorization, etc.
- Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



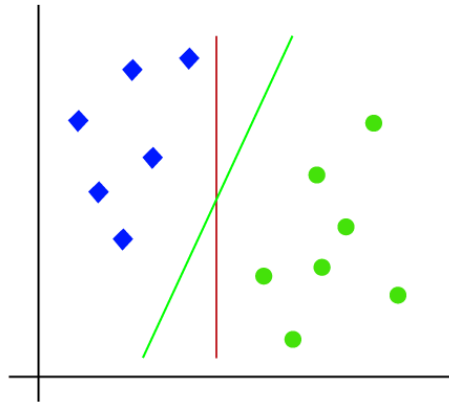
Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



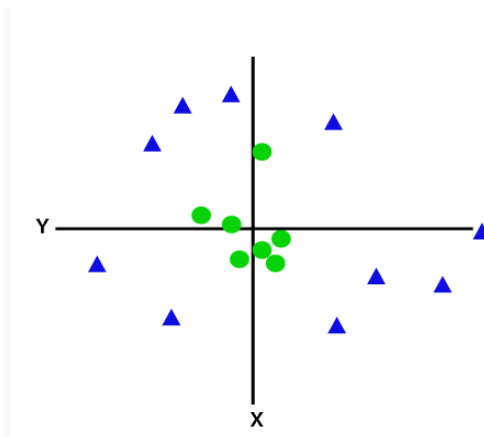
Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.



- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.



Prediction:

- Prediction is a fundamental task in data mining that involves using historical data to forecast unknown or future values.
- Unlike descriptive analysis, which summarizes past data, prediction aims to infer new outcomes. For example, based on past customer purchases, a retailer might predict which products a customer is likely to buy next.
- In data mining, prediction typically refers to estimating a target variable's value, which can be either categorical (classification) or continuous (regression).
- It helps convert vast amounts of raw data into meaningful forecasts that support decision-making processes.
- For example: Predicting the worth of a home based on facts like the number of rooms, total area, and so on.

Accuracy and Error Measures:

1) Accuracy:

Predictive Accuracy describes the degree to which predicted values match the actual values of a target field, accounting for statistical noise and fluctuations.

Accuracy is one metric for Evaluating Classification model. Informally, accuracy is the function of prediction our model got right.

Accuracy=Number of correct predictions/ total Number of prediction

For binary Classification, accuracy can also be calculated in terms of positive and negative as follows

Accuracy=TP+TN/Total

2) Error Measures:

This refers to the discrepancy between recorded data values and the actual "true" values, often resulting from the measurement process.

In other words, the recorded data values differ from true values to some extent. The difference between measured and true value is called the error.

The Difference between measured and true value is called error.

Error Measures=1-Acuuracy

Error Measures=FN+FT/Total

Evaluating the Accuracy of a Classifier or Predictor:

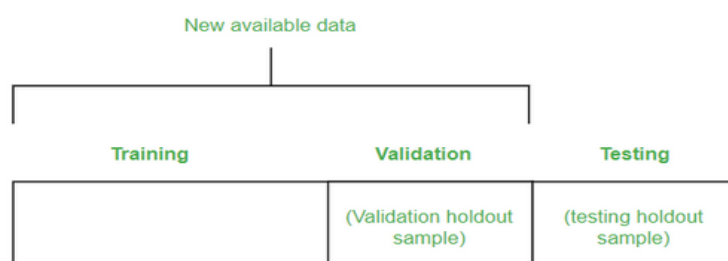
Techniques to Evaluate Accuracy of Classifier in Data Mining

1) HoldOut:

In the holdout method, the largest dataset is randomly divided into three subsets:

1. A training set is a subset of the dataset which are been used to build predictive models.
2. The validation set is a subset of the dataset which is been used to assess the performance of the model built in the training phase.
3. Test sets or unseen examples are the subset of the dataset to assess the likely future performance of the model.

Basically, two-thirds of the data are being allocated to the training set and the remaining one-third is being allocated to the test set.



2) Random Subsampling:

Random subsampling is a variation of the holdout method. The holdout method is being repeated K times.

The holdout subsampling involves randomly splitting the data into a training set and a test set.

On the training set the data is being trained and the mean square error (MSE) is been obtained from the predictions on the test set.

As MSE is dependent on the split, this method is not recommended. So a new split can give you a new MSE.

3) Cross-Validation:

K-fold cross-validation is being used when there is only a limited amount of data available, to achieve an unbiased estimation of the performance of the model.

Here, we divide the data into K subsets of equal sizes.

We build models K times, each time leaving out one of the subsets from the training, and use it as the test set.

If K equals the sample size, then this is called a "Leave-One-Out".

4) Bootstrapping:

Bootstrapping is one of the techniques which is used to make the estimations from the data by taking an average of the estimates from smaller data samples.

The bootstrapping method involves the iterative resampling of a dataset with replacement.

On resampling instead of only estimating the statistics once on complete data, we can do it many times.

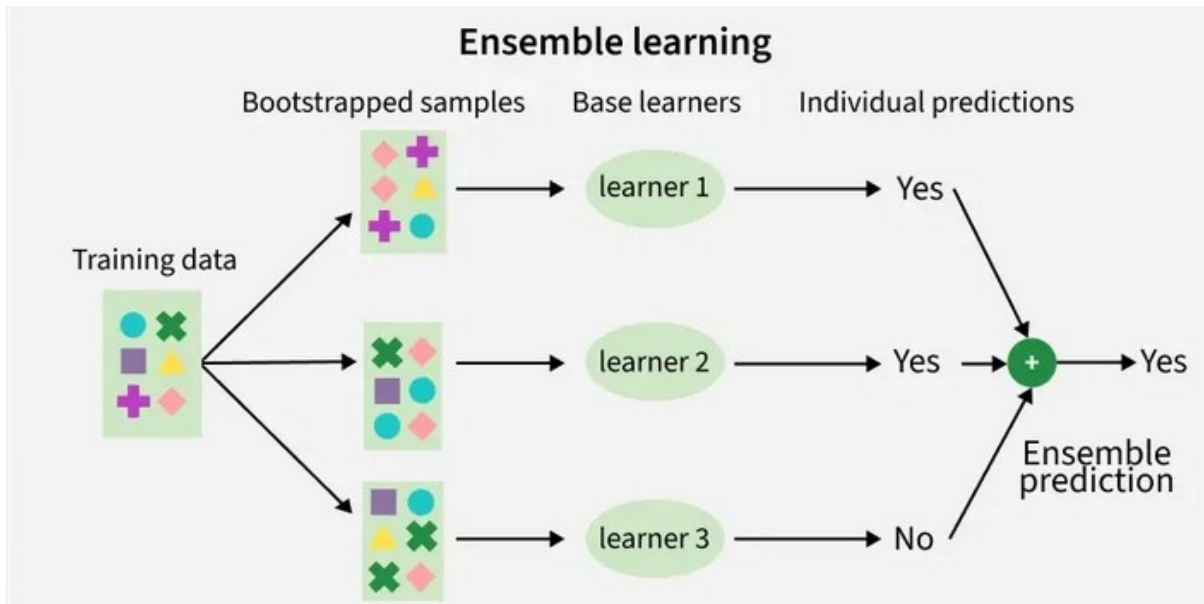
Repeating this multiple times helps to obtain a vector of estimates.

Bootstrapping can compute variance, expected value, and other relevant statistics of these estimates.

Ensemble Methods:

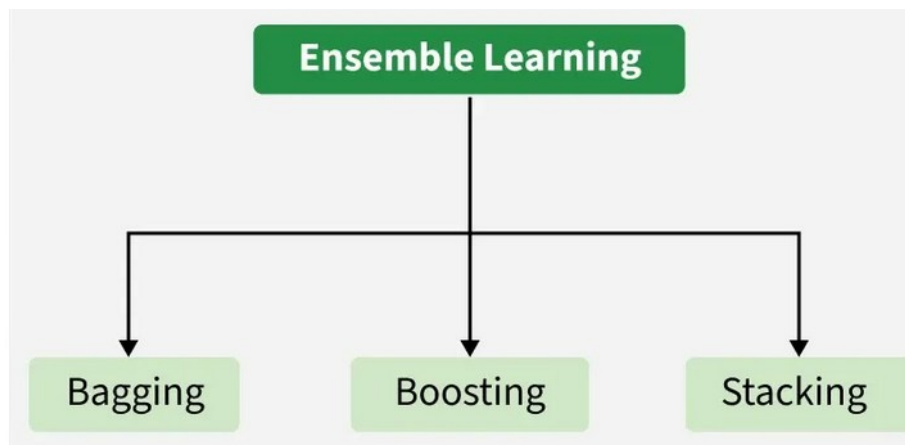
- Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.
- Ensemble learning improves machine learning accuracy and robustness by combining multiple "weak" base models to create a single "strong" model.
- In simple words, ensemble methods in machine learning use more than one model to make better predictions.
- Think of it like teamwork. One model may make mistakes, but a group of models can correct each other and give stronger results.
- **Multiple Models:** Uses many small models together
- **Better Accuracy:** Combined results improve performance

- **Reduced Errors:** Mistakes of one model are balanced by those of others
- **Simple Idea:** Like taking advice from a group instead of one person



Types of Ensembles Learning

There are three main types of ensemble methods:



1. Bagging (Bootstrap Aggregating):

- Models are trained independently on different random subsets of the training data. Their results are then combined usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.
- Runs multiple models in parallel independently, often using decision trees on subsets of data, and averages results (e.g., Random Forest).

Bagging

Different Subset
of the Training Data



Base Models



Most Frequent or
Average Prediction



Prediction

2. Boosting:

- Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones.
- The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.
- Trains models sequentially, where each new model attempts to correct the errors of the previous one.

Boosting

Base Models



Prediction



Prediction

The training is
modified based on
the predictions

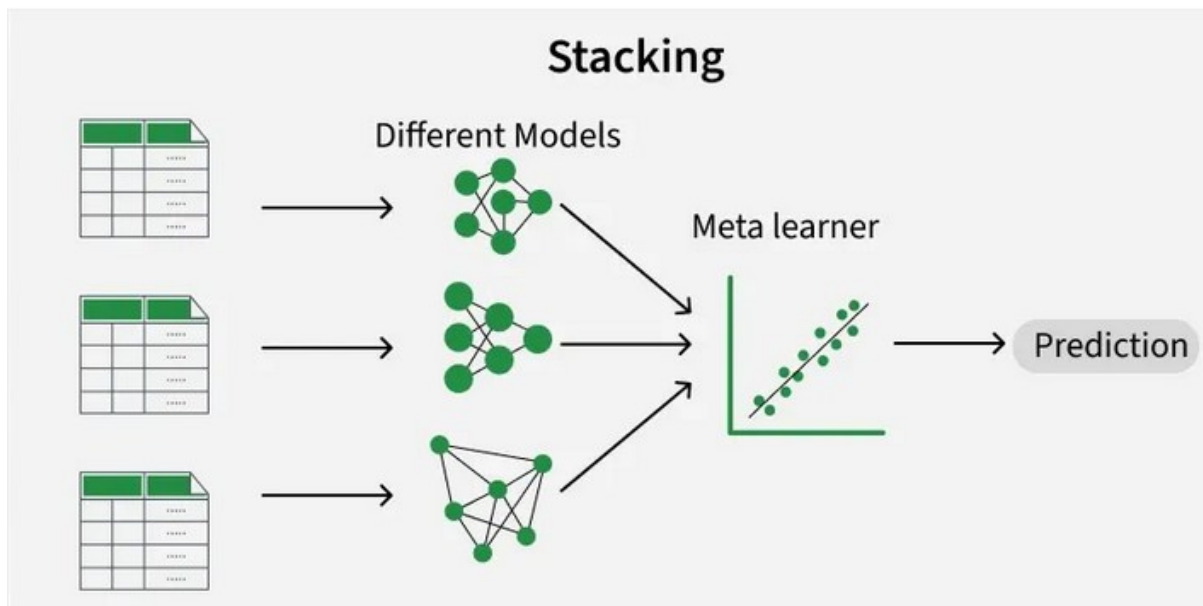


Prediction

3. Stacking (Stacked Generalization):

- Multiple different models (often of different types) are trained and their predictions are used as inputs to a final model, called a meta-model.
- The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.

- Combines multiple heterogeneous base models (e.g., SVM, Decision Tree, Logistic Regression) through a meta-model that learns the best way to combine them.



Lazy Learner:

- The K-Nearest Neighbors (KNN) algorithm is known as a lazy learner
- During the training phase of lazy learning the algorithm accepts the data as input but refrains from actively training on it.
- lazy learner because it does not build an internal model during the training phase. Instead, it simply stores the entire training dataset and defers any processing until it needs to make a prediction.
- Lazy learning algorithms like K-NN are useful when the data is noisy or has a complex structure.

Here's how K-NN works:

- **Training Phase:** K-NN doesn't actually "train" in the traditional sense—it just stores the training data.
- **Classification Phase:** When new data needs to be classified, K-NN finds the K closest points in the training data to the new point and assigns it a label based on the majority label among those nearest neighbors.

However, K-NN can be slow for large datasets because it has to search through all the data points each time it makes a prediction. Unlike eager learning algorithms, which build a model during training, lazy learning algorithms simply store the data, making their training phase extremely quick since no processing or learning happens at that stage.